

Monologue on OPERATING SYSTEMS (UNIT 3)

Step 1. Introduction 1. Start with a hook sentence that will attract the listener's attention, a quote, a proverb, etc. 2. Lead your speech steadily to the main part of your talk. 3. The introduction may consist of 3-6 sentences.	The role of operating systems in modern computing is pivotal, as they serve as the backbone of every computer system. As Linus Torvalds once said, "An operating system never does anything on its own; it's only waiting for the programs to ask for certain resources." This quote highlights the fundamental role of an operating system as an intermediary between hardware and software. Linus Torvalds, the creator of Linux, emphasizes that an operating system's primary mission is to facilitate the execution of programs by providing necessary resources and services.
Step 2. How do computers work? 2.1. What is an OS? 2.2. Types of OS (GUI, Multi-user, Multiprocessing, Multithreading, Multitasking). What are they?	An operating system (OS) is a software that acts as an interface between computer hardware components and the user. It manages all other application programs in a computer, enabling them to run efficiently by providing essential services like resource management and hardware abstraction . The OS helps users communicate with the computer without needing to understand its language, making it indispensable for running applications like browsers, MS Office, and games. Operating systems can be categorized into several types based on their functionality. For example, GUI (Graphical User Interface) operating systems use graphics and icons to provide an intuitive interface. Examples include Windows and macOS. Multi-user operating systems allow multiple users to use the same computer at the same time and at different times , which is common in server environments. Multiprocessing operating systems support and utilize more than one computer processor , enhancing performance. Multitasking operating systems allow multiple software processes to run at the same time , improving productivity. Multithreading operating systems allow different parts of a software program to run concurrently , optimizing resource usage. These types of operating systems enable computers to efficiently manage resources , enhance user experience , and optimize performance .
Step 3. What do OS do? 3.1. Why is an OS one of the most important parts of a computer? 3.2. Speak about the tasks operating systems must accomplish.	An operating system is one of the most important parts of a computer because it controls the computer's resources , ensuring that applications run smoothly. It provides adaptability towards different uses , making it versatile for various tasks. Operating systems must accomplish several critical tasks: they allocate resources such as CPU time and memory to ensure applications run efficiently . Resources are allocated by the operating system to each process, and this allocation is managed to prevent crashes and errors. They provide a consistent application program interface (API) , allowing developers to create compatible software. They ensure that each process has enough memory to execute , preventing crashes and errors. They translate electrical signals to hardware devices , facilitating communication between software and hardware. These tasks are crucial for maintaining system stability and performance, allowing multiple applications to run simultaneously and balancing resource needs .
Step 4. History of OS 4.1. Speak about the generations of operating systems.	The evolution of operating systems spans four generations. The first generation (1945-1955) saw no operating systems; programming was done using machine language and plugboards . The second generation (1955-1965) introduced batch processing systems where jobs were

4.2. Compare two modern OS.	submitted in batches. The third generation (1965-1980) saw the development of multiprogramming techniques and time-sharing systems. The fourth generation (1980-present) is characterized by personal computers and modern operating systems like Windows and macOS. Comparing two modern operating systems, Windows and Linux: Windows is widely used for its user-friendly interface and compatibility with a wide range of applications. Linux is preferred for its security, flexibility, and open-source nature, making it ideal for servers and developers. Both systems take better advantage of computer resources and support multitasking, but they differ in cost and user interface complexity. If operating systems did not exist, computers would be unable to run applications effectively, and software development would be impractical. Therefore, people say it is crucial that operating systems continue to evolve to meet the demands of emerging technologies.
Step 5. CREATIVE THINKING Introduce your own extra idea(s) on advertising that hasn't/haven't been mentioned before. Substantiate your choice.	In the future, operating systems will likely become even more sophisticated, enabling computers to perform tasks more efficiently. For instance, future operating systems might integrate AI to optimize resource allocation, further enhancing system performance.
Step 6. Conclusion Summarise the ideas of steps 2,3,4,5.	In conclusion, operating systems are vital for modern computing, providing essential services that enable computers to function efficiently. They manage resources, facilitate multitasking, and offer a user interface, making them indispensable for both personal and enterprise environments.

Active Vocabulary: 30, **Grammar Structures:** 6, **Linkers:** 13. Total: 654 words